The Network Layer: Data Plane & Control Plane

Mechanics of Packet Switching and Routing

Bernal Manzanilla

Concordia University of Edmonton
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The Network Layer: Two Interacting Planes

The primary role of the network layer is to move packets from a sending host to a receiving host. This can be decomposed into two interacting parts:

- **Data Plane:** The primary data-plane role of each router is to forward datagrams from its input links to its output links.
- **Control Plane:** The primary role of the network control plane is to coordinate these local, per-router forwarding actions so that datagrams are ultimately transferred end-to-end, along paths of routers between source and destination hosts.

Defining Forwarding and Routing

The terms forwarding and routing are often used interchangeably, but there is a precise and important distinction:

Forwarding (Data Plane)

Forwarding refers to the router-local action of transferring a packet from an input link interface to the appropriate output link interface[cite: 704]. Forwarding takes place at very short timescales (typically a few nanoseconds), and thus is typically implemented in hardware.

Routing (Control Plane)

Routing refers to the network-wide process that determines the end-to-end paths that packets take from source to destination[cite: 706]. Routing takes place on much longer timescales (typically seconds), and is often implemented in software.

What's Inside a Router?

[Image of router architecture]

A typical router consists of four main components:

- **Input Ports:** Performs the physical layer function of terminating an incoming physical link and the link-layer functions needed to interoperate with the other side of the link. Crucially, a lookup function is performed here, where the forwarding table is consulted to determine the output port.
- **Switching Fabric:** The switching fabric connects the router's input ports to its output ports.
- **Output Ports:** An output port stores packets received from the switching fabric and transmits these packets on the outgoing link by performing the necessary link-layer and physical-layer functions.
- **Routing Processor:** The routing processor performs control-plane functions, executing routing protocols to compute the forwarding table or communicating with a remote SDN controller.

Input Port Processing: Longest Prefix Matching

To handle the scale of 4 billion IPv4 addresses, routers match a prefix of the packet's destination address with the entries in the table.

The Longest Prefix Matching Rule

When a destination address matches multiple entries in the forwarding table, the router uses the longest matching prefix; that is, it finds the longest matching entry in the table and forwards the packet to the link interface associated with the longest prefix match.

At Gigabit transmission rates, this lookup must be performed in nanoseconds in hardware, typically using Ternary Content Addressable Memories (TCAMs).

The Switching Fabric

The switching fabric is completely contained within the router—a network inside of a network router[cite: 856]. Switching can be accomplished in a number of ways:

- **Switching via memory:** The packet is copied from the input port into processor memory, and the processor copies the packet to the output port's buffers.
- **Switching via a bus:** An input port transfers a packet directly to the output port over a shared bus[cite: 996]. Because every packet must cross the single bus, the switching speed of the router is limited to the bus speed[cite: 1001].
- **Switching via an interconnection network:** A crossbar switch is an interconnection network consisting of $2N$ buses that connect N input ports to N output ports[cite: 1006]. A crossbar switch is non-blocking—a packet being forwarded to an output port will not be blocked as long as no other packet is currently being forwarded to that output port.

Queuing and Packet Loss

The location and extent of queuing will depend on the traffic load, the relative speed of the switching fabric, and the line speed. As queues grow large, the router's memory can eventually be exhausted and packet loss will occur.

- **Head-of-the-Line (HOL) Blocking:** A phenomenon in an input-queued switch where a queued packet in an input queue must wait for transfer through the fabric (even though its output port is free) because it is blocked by another packet at the head of the line.
- **Bufferbloat:** A scenario of long delay due to persistent buffering, occurring when the end-to-end pipe is full but the amount of queuing delay is constant and persistent.

The IPv4 Datagram

The IPv4 datagram format contains several key fields used by the network layer:

- **Time-to-live (TTL):** The TTL field is included to ensure that datagrams do not circulate forever in the network. This field is decremented by one each time the datagram is processed by a router. If the TTL field reaches 0, a router must drop that datagram.
- **Protocol:** The value of this field indicates the specific transport-layer protocol to which the data portion of this IP datagram should be passed (e.g., a value of 6 indicates TCP, while 17 indicates UDP).
- **Destination IP Address:** Used by the router to index into its forwarding table to determine the outgoing link interface.

Network Address Translation (NAT)

NAT allows small office/home office (SOHO) networks to operate using a single allocated IP address.

- **Private Network Realm:** A network whose addresses only have meaning to devices within that network (e.g., the 10.0.0.0/8 address space).
- **The NAT Router:** The NAT-enabled router behaves to the outside world as a single device with a single IP address, hiding the details of the home network from the outside world.
- **NAT Translation Table:** The router uses this table to rewrite the source IP address and source port number for outgoing traffic, and reverse the process for incoming traffic.

The Control Plane: Two Architectures

There are two possible approaches for computing, maintaining, and installing forwarding tables:

1. Per-Router Control

A routing algorithm runs in each and every router; both a forwarding and a routing function are contained within each router. Each router's routing component communicates with the routing components in other routers to compute the values for its forwarding table (e.g., OSPF and BGP).

2. Logically Centralized Control (SDN)

A logically centralized controller computes and distributes the forwarding tables to be used by each and every router. The controller interacts with a control agent (CA) in each of the routers via a well-defined protocol.

Routing Algorithms: Centralized vs. Decentralized

A natural goal of a routing algorithm is to identify the least costly paths between sources and destinations.

- **Centralized Routing Algorithm:** Computes the least-cost path between a source and destination using complete, global knowledge about the network. The algorithm takes the connectivity between all nodes and all link costs as inputs. Algorithms with global state information are often referred to as **link-state (LS) algorithms**[cite: 2557, 2558, 2562].
- **Decentralized Routing Algorithm:** The calculation of the least-cost path is carried out in an iterative, distributed manner by the routers. No node has complete information about the costs of all network links. This is called a **distance-vector (DV) algorithm**.

Distance-Vector (DV) Routing & Count-to-Infinity

The DV algorithm relies on the **Bellman-Ford equation**: $d_x(y) = min_v\{c(x,v) + d_v(y)\}$. Each node maintains a vector of estimates of the costs to all other nodes, updating it when it receives distance vectors from its neighbors.

The Count-to-Infinity Problem

When link costs increase, a routing loop can occur. A packet destined for node X arriving at Y or Z will bounce back and forth between these two nodes forever (or until the forwarding tables are changed). The bad news about the increase in link cost travels slowly.

Scaling Routing: Autonomous Systems

We cannot run a single routing algorithm across the entire Internet due to scale (hundreds of millions of routers) and administrative autonomy (ISPs desire to operate their networks as they please).

Both of these problems can be solved by organizing routers into **Autonomous Systems (ASs)**, with each AS consisting of a group of routers that are under the same administrative control.

- **Intra-AS Routing:** The routing algorithm running within an autonomous system (e.g., OSPF).
- **Inter-AS Routing:** A routing protocol that involves coordination among multiple ASs to route a packet across multiple ASs (e.g., BGP).

Intra-AS Routing: OSPF

OSPF (Open Shortest Path First):

- OSPF is a link-state protocol that uses flooding of link-state information and a Dijkstra's least-cost path algorithm.
- With OSPF, each router constructs a complete topological map (that is, a graph) of the entire autonomous system.
- Each router then locally runs Dijkstra's shortest-path algorithm to determine a shortest-path tree to all subnets, with itself as the root node.

Inter-AS Routing: BGP

BGP (Border Gateway Protocol):

- BGP is the protocol that glues the thousands of ISPs in the Internet together[cite: 3080].
- In BGP, packets are not routed to a specific destination address, but instead to CIDRized prefixes, with each prefix representing a subnet or a collection of subnets.
- BGP allows each subnet to advertise its existence to the rest of the Internet. If it weren't for BGP, each subnet would be an isolated island.

BGP: Hot Potato Routing

When a router learns about multiple paths to a given destination, how does it choose?

Hot Potato Routing

In hot potato routing, the route chosen (from among all possible routes) is that route with the least cost to the NEXT-HOP router beginning that route.

The idea behind hot potato routing is for a router to get packets out of its AS as quickly as possible (with the least cost possible) without worrying about the cost of the remaining portions of the path outside of its AS. It is a selfish algorithm.

The Internet Control Message Protocol (ICMP)

- ICMP is used by hosts and routers to communicate network-layer information to each other. The most typical use of ICMP is for error reporting.
- **Ping:** The ping program sends an ICMP type 8 code 0 message to the specified host. The destination host sends back a type 0 code 0 ICMP echo reply.
- **Traceroute:** Traceroute sends UDP segments with increasing TTL values. When the TTL expires at an intermediate router, the router discards the datagram and sends an ICMP warning message (type 11 code 0) back to the source.